# Lecture 20: TCP Part 2

## Congestion Control and the Modern Picture

EC 441 — Introduction to Computer Networking
Boston University

Spring 2026

# Learning Objectives

By the end of this lecture, you should be able to:

- Explain why TCP congestion control is necessary and differs from flow control
- Describe slow start, congestion avoidance (AIMD), and fast recovery
- Trace **cwnd** and **ssthresh** evolution in response to timeout and triple dup ACK
- State the TCP throughput formula and explain its RTT/loss-rate dependence
- Explain why high-BDP paths need window scaling and why Reno fails there
- Describe CUBIC's key departure from Reno
- Explain ECN and why it is preferable to loss-based signals
- Describe the AIMD fairness argument
- Explain why QUIC was designed and what TCP limitations it addresses

# Key Acronyms

| Acronym | Meaning |
|---|---|
| ACK | Acknowledgment |
| AIMD | Additive Increase Multiplicative Decrease |
| BDP | Bandwidth-Delay Product |
| CA | Congestion Avoidance |
| cwnd | Congestion Window |
| ECN | Explicit Congestion Notification |

| Acronym | Meaning |
|---|---|
| HOL | Head-of-Line (blocking) |
| MSS | Maximum Segment Size |
| RTT | Round-Trip Time |
| RTO | Retransmission Timeout |
| ssthresh | Slow-Start Threshold |
| QUIC | Quick UDP Internet Connections |

# Callback — The Missing Piece from L19

**L19 left one variable undefined:** `cwnd`

| Limit | Set by | Protects |
|---|---|---|
| `rwnd` | Receiver | Receiver's buffer |
| `cwnd` | Sender | Network from congestion |
| Effective window | min(`cwnd`, `rwnd`) — both | |

- **Flow control** is *explicit*: receiver tells sender its available buffer via `rwnd`
- **Congestion control** is *implicit*: sender infers network state from loss signals
- *No router ever says "I am congested" — TCP must infer it from dropped packets.*

# Why Congestion Control?

- **Congestion collapse**: aggregate traffic exceeds capacity $\Rightarrow$ queues fill $\Rightarrow$ drops $\Rightarrow$ retransmits $\Rightarrow$ even more traffic $\Rightarrow$ collapse
- **TCP's contract**: probe aggressively for bandwidth; back off immediately and sharply on loss
- Without self-regulation: greedy senders starve all others
- TCP Reno uses loss as the congestion signal:
  - **Timeout** — severe signal (nothing getting through)
  - **3 duplicate ACKs** — mild signal (later packets still arriving)

*"No router ever says 'I am congested' — TCP must infer it."*

# Slow Start

- `cwnd` starts at **1 MSS**; increases by 1 MSS per ACK received $\Rightarrow$ doubles every RTT (exponential growth)
- "Slow" refers to the starting point (1 MSS), *not* the growth rate
- **ssthresh** controls when slow start stops:
  - `cwnd` $<$ `ssthresh`: slow start (exponential)
  - `cwnd` $\geq$ `ssthresh`: congestion avoidance (linear, $+1$ MSS/RTT)
- `ssthresh` initialized large; set to `cwnd`/2 on first loss

| RTT | cwnd (MSS) |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 4 |
| 3 | 8 |
| 4 | 16 |

Each ACK $\Rightarrow +1$ MSS
Each RTT $\Rightarrow$ cwnd $\times 2$

# Congestion Avoidance (AIMD) and the Sawtooth

**Additive Increase** (CA phase):

`cwnd` $+= \frac{\text{MSS}^2}{\texttt{cwnd}}$ per ACK

$\approx +1$ MSS per RTT

**On 3 dup ACKs (mild loss):**

`ssthresh` $=$ `cwnd`$/2$

`cwnd` $=$ `ssthresh`

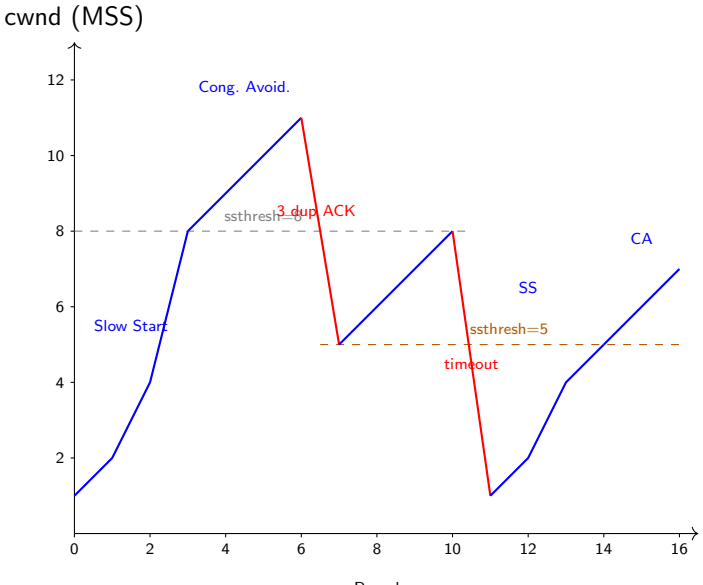
(fast recovery; skip slow start)

**On timeout (severe loss):**

`ssthresh` $=$ `cwnd`$/2$

`cwnd` $= 1$ MSS

Restart slow start



# Fast Recovery

When **3 duplicate ACKs** arrive: fast retransmit + fast recovery

1. `ssthresh` = `cwnd` /2
2. `cwnd` = `ssthresh` *(skip slow start — dup ACKs prove later packets flow)*
3. Retransmit the missing segment immediately
4. Each *additional* dup ACK while in recovery: `cwnd` + = 1 MSS (each dup ACK = one more segment delivered to receiver)
5. New ACK arrives: `cwnd` = `ssthresh` ⇒ return to congestion avoidance

**Fast recovery (3 dup ACKs):**
Network partly working; reduce cwnd by half, stay in CA

**Timeout:**
Nothing getting through; `cwnd` → 1, full slow start

# Worked cwnd Trace

| Event | cwnd before | Action | cwnd after | ssthresh |
|---|---|---|---|---|
| Start | — | SS begins | 1 | 8 |
| ACK (SS) | 1 | $\times 2$/RTT | 2 | 8 |
| ACK (SS) | 2 | $\times 2$/RTT | 4 | 8 |
| ACK (SS) | 4 | $\times 2$/RTT | 8 | 8 |
| ACK (CA) | 8 | $+1$/RTT | 9 | 8 |
| ACK (CA) | 9 | $+1$/RTT | 10 | 8 |
| 3 dup ACKs | 10 | ssthresh=5, cwnd=5 | 5 | 5 |
| New ACK | 5 | CA resumes | 6 | 5 |
| timeout | 8 | ssthresh=4, cwnd=1 | 1 | 4 |
| SS resumes | 1 | exponential $\rightarrow$ 4 | ... | 4 |

At **ssthresh** the mode switches: slow start (exponential) $\rightarrow$ congestion avoidance (linear).
Timeout resets `cwnd` to 1; 3 dup ACKs halve `cwnd` (much gentler).

# Demo — ss -ti

**Reading TCP internals directly from the kernel:**

```
$ ss -ti dst 93.184.216.34
tcp ESTAB 0 0 192.168.1.5:45678 93.184.216.34:443
     cubic wscale:7,7 rto:204 rtt:4.5/1.1 ato:40 mss:1448
     cwnd:10 ssthresh:10
     send 257.6Mb/s lastsnd:8 lastrcv:8 lastack:8
     app_limited:1
```

| | |
|---|---|
| `cubic` | Congestion control algorithm (Linux default since kernel 2.6.19) |
| `rtt:4.5/1.1` | SRTT = 4.5 ms, RTTVAR = 1.1 ms (Jacobson estimator from L19) |
| `rto:204` | RTO formula $\approx$ 9 ms; RFC 6298 floor = 200 ms — floor is active |
| `cwnd:10 ssthresh:10` | At the CA/SS boundary |
| `app_limited:1` | Application, not network, is the bottleneck here |

# TCP Throughput Formula

## Mathis et al. (1997) steady-state approximation

$$\text{BW} \approx \frac{\text{MSS}}{\text{RTT}} \cdot \frac{1}{\sqrt{p}}$$

- **Intuition**: sawtooth between $W/2$ and $W$; average cwnd $\approx 3W/4$; loss rate $p \approx 8/(3W^2)$
- Throughput $\propto 1/\text{RTT}$ — longer paths get systematically less bandwidth
- Throughput $\propto 1/\sqrt{p}$ — loss is very costly

## Striking implication

To achieve **10 Gb/s** over RTT = 100 ms with MSS = 1460 B, the required loss rate is $p < 10^{-10}$ — fewer than one dropped segment per 10 billion sent. In practice, high-speed TCP needs *near-zero* loss.

# Bandwidth-Delay Product and Window Scaling

**BDP** $= \text{BW} \times \text{RTT} =$ "pipe capacity" in bytes

**Problem:** TCP Window field $= 16$ bits $\Rightarrow$ max **64 KB** without options

Example: 1 Gb/s $\times$ 100 ms $= 12.5$ MB needed; $\frac{64\ \text{KB}}{0.1\ \text{s}} = 5.2$ Mb/s ($< 1\%$ of link!)

**Window Scale option** (RFC 7323): multiplies window by $2^n$, $n \in [0, 14] \Rightarrow$ up to 1 GB

| BW | RTT | BDP | Required wscale |
|---:|---:|---:|---:|
| 100 Mb/s | 10 ms | 125 KB | 1 ($\times$2) |
| 1 Gb/s | 100 ms | 12.5 MB | 8 ($\times$256) |
| 10 Gb/s | 100 ms | 125 MB | 12 ($\times$4096) |

# TCP Reno vs. TCP CUBIC

| | **TCP Reno** | **TCP CUBIC** |
|---|---|---|
| Growth clock | RTT-clocked | Wall-clock time |
| Growth shape | Linear (AIMD) | Cubic function of time since last loss |
| High-BDP behavior | Slow recovery | Fast recovery (large $W$ far from last loss) |
| RTT fairness | Short RTT flows get more | RTT-independent growth |
| Linux default | No (superseded) | Yes (since kernel 2.6.19) |

## CUBIC key idea

$W(t) = C\,(t - K)^3 + W_{\max}$ where $K$ is chosen so that $W(0) = W_{\max}/2$. Growth is slow near the last loss point (probing cautiously) and fast when far from it (filling available bandwidth quickly).

Note: CUBIC is the default congestion control in Linux, macOS, and most modern OSes.

# ECN — Congestion Without Dropping

**Loss-based signaling (today's default):**

1. Router's queue fills
2. Router drops packet
3. ACK gap detected by sender
4. `cwnd` reduced — *after* losing goodput

**ECN (RFC 3168):**

1. Router's queue *begins to build*
2. Router **marks** IP header (CE bit) — packet delivered!
3. Receiver echoes CE via **ECE** flag in TCP ACK
4. Sender reduces `cwnd` as if loss occurred

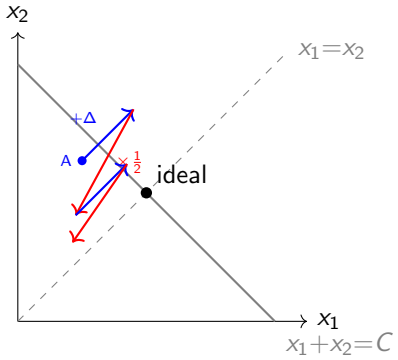
**Benefits**

- No goodput lost from the dropped packet
- Earlier signal (queue building, not yet overflowing)
- Lower latency under load

**Caveat**

Requires both endpoints *and* all routers to support ECN. Widely deployed in datacenters; less universal on the public Internet.

# AIMD Fairness




Two flows sharing capacity $C$:

- **Additive increase**: both grow by $\Delta$ per RTT — move *parallel* to fairness line
- **Multiplicative decrease**: both halve — move toward *origin*
- Net effect: spiral toward equal-share, full-utilization point

# Caveats on AIMD Fairness

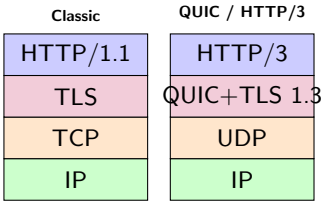
- **RTT unfairness**: TCP Reno gives more bandwidth to shorter-RTT flows (more AIMD steps per second). CUBIC partially fixes this — wall-clock growth is RTT-independent.
- **UDP unfriendliness**: UDP flows do not participate in AIMD. A greedy UDP sender will crowd out TCP. "Friendly" protocols (DCCP, QUIC) implement their own rate control.
- **Multiple connections**: a client opening $N$ TCP connections to a server captures $N$ times the AIMD share. HTTP/1.1 browsers exploited this with 6 parallel connections per host.

# What TCP Gets Wrong (and Why It's Hard to Fix)

1. **HOL blocking in multiplexed streams** — HTTP/2 runs many streams over one TCP byte-stream; one lost packet *blocks all streams* until retransmit. QUIC delivers each stream independently.
2. **Slow connection setup** — TCP 3WHS + TLS handshake = 2–3 RTTs before data flows. QUIC + TLS 1.3: **1 RTT** new connection, **0 RTT** resumed.
3. **Ossification** — Middleboxes inspect and modify TCP headers; changing TCP semantics is nearly impossible in practice. QUIC header is *fully encrypted* (over UDP); middleboxes cannot see or tamper with it.
4. **No connection migration** — TCP is identified by a 4-tuple; switching networks (e.g., WiFi → cellular) breaks the connection. QUIC uses a **connection ID** independent of addresses.

*The fix requires updating all deployed middleboxes — effectively impossible. QUIC bypasses the problem by running over UDP.*

# QUIC — What It Is




**QUIC replaces TCP + TLS for web traffic:**

- Transport protocol over UDP
- Mandatory built-in TLS 1.3
- Multiple *independent* streams (no HOL blocking)
- Own congestion control (CUBIC or BBR)
- Used by: HTTP/3, YouTube, Google Search
- $\approx$ 25% of Internet traffic (2024)

# UDP / TCP / QUIC at a Glance

| Feature | UDP | TCP | QUIC |
|---|---|---|---|
| Connection setup | None | 3-way handshake (1 RTT) | 1 RTT (0 RTT resumed) |
| Reliability | None | Yes (cumul. ACK + retx) | Yes (per stream) |
| Ordering | None | Yes (byte stream) | Yes (per stream) |
| Flow control | None | rwnd | Yes |
| Congestion control | None | AIMD (cwnd) | CUBIC / BBR |
| HOL blocking | N/A | Yes (one byte stream) | No (indep. streams) |
| Encryption | None (app layer) | TLS separate | TLS 1.3 built-in |
| Header size | 8 bytes | 20+ bytes | Variable (encrypted) |
| Runs over | IP | IP | UDP |
| Typical use | DNS, streaming, games | Web, SSH, bulk transfer | HTTP/3, YouTube |

*QUIC combines the low overhead of UDP with the reliability of TCP and fixes TCP's structural limitations.*

# The Socket Programmer's View

**The application sees**: a connected socket — reliable, ordered, full-duplex byte stream. All TCP machinery (RTT estimation, cwnd, retransmits) is *invisible to the app*.

`send()` writes bytes in; `recv()` reads bytes out.

| Socket Option | Effect |
|---|---|
| `TCP_NODELAY` | Disable Nagle's algorithm; send immediately (low latency) |
| `TCP_INFO` (Linux) | Read cwnd, ssthresh, RTT, retransmits from kernel |
| `SO_RCVBUF` / `SO_SNDBUF` | Tune socket buffer sizes; affects rwnd and throughput |

"This is what you will use in L24 (sockets lab)."

# Transport Layer Summary — L18–L20

| Concept | Where | Key idea |
|---|---|---|
| Multiplexing / demux | L18 | Port numbers identify processes |
| UDP | L18 | Best-effort, 8-byte header, app controls everything |
| Stop-and-Wait / GBN / SR | L18 | Pipeline mechanisms; S&W utilization, window fills pipe |
| TCP segment header | L19 | 20-byte min; seq/ack are byte offsets |
| 3-way handshake | L19 | SYN/SYN-ACK/ACK; ISN randomization |
| Reliable delivery + RTO | L19 | EWMA + RTTVAR; Karn's algorithm; backoff |
| Flow control (rwnd) | L19 | Receiver-driven; prevents buffer overflow |
| Slow start / AIMD | L20 | Probe then back off; cwnd doubles, then +1/RTT |
| Fast recovery | L20 | 3 dup ACKs $\rightarrow$ skip slow start |
| Congestion window | L20 | Effective window = min(cwnd, rwnd) |
| QUIC | L20 | UDP-based; fixes HOL, ossification, migration |

# Discussion Questions

1. Why does TCP Reno perform poorly on satellite links (high RTT, high BDP) even at low loss rates?
2. Two TCP flows share a 100 Mb/s link: one has RTT = 10 ms, the other RTT = 100 ms. Do they share the bandwidth equally under Reno? How does CUBIC improve this?
3. If a router uses ECN instead of drops, what happens to the TCP throughput formula? Does $p$ still measure the same thing?
4. A UDP video application has no rate control. What happens to co-existing TCP flows? Is this fair?
5. HTTP/2 claims to solve HOL blocking with multiplexing. Why does QUIC claim HTTP/2 over TCP *still* has HOL blocking?
6. Why must QUIC implement its own congestion control rather than relying on the OS TCP stack?